

Knowledge Bases

Lecture 11: Modal Logics II

dr Anna Maria Radzikowska

Wydział Matematyki i Nauk Informacyjnych

Pokój 504, gmach MiNI

E-mail: A.Radzikowska@mini.pw.edu.pl

Propositional Temporal Logic (PDL)

Temporal logic — modal logic of time (Prior, 1957). Modal operators $\Box$ and $\Diamond$ are interpreted here as “*always in the future*” and “*sometime in the future*”, respectively.

Temporal logic — modal logic of time (Prior, 1957). Modal operators $\Box$ and $\Diamond$ are interpreted here as “*always in the future*” and “*sometime in the future*”, respectively.

The alphabet of the language of PDL consists of:

- denumerable set of propositional variables Var
- logical connectives $\neg, \rightarrow$
- modal operators: G (“*going to be*”), F (“*future*”), H (“*has been*”), P (“*past*”), $\bigcirc$ (“*next time*”), and $*$ (“*previous time*”).

Intuitive interpretation

Intuitively, modal formulas are read as:

● $G\alpha$: *always in the future α holds*

Intuitive interpretation

Intuitively, modal formulas are read as:

- $G\alpha$: *always in the future α holds*
- $F\alpha$: *at some time in the future α holds*

Intuitive interpretation

Intuitively, modal formulas are read as:

- $G\alpha$: *always in the future α holds*
- $F\alpha$: *at some time in the future α holds*
- $H\alpha$: *always in the past α was satisfied*

Intuitive interpretation

Intuitively, modal formulas are read as:

- $G\alpha$: *always in the future α holds*
- $F\alpha$: *at some time in the future α holds*
- $H\alpha$: *always in the past α was satisfied*
- $P\alpha$: *at some moment in the past α was satisfied*

Intuitive interpretation

Intuitively, modal formulas are read as:

- $G\alpha$: *always in the future α holds*
- $F\alpha$: *at some time in the future α holds*
- $H\alpha$: *always in the past α was satisfied*
- $P\alpha$: *at some moment in the past α was satisfied*
- $\bigcirc\alpha$: *next moment α will be satisfied*

Intuitive interpretation

Intuitively, modal formulas are read as:

- $G\alpha$: *always in the future α holds*
- $F\alpha$: *at some time in the future α holds*
- $H\alpha$: *always in the past α was satisfied*
- $P\alpha$: *at some moment in the past α was satisfied*
- $\bigcirc\alpha$: *next moment α will be satisfied*
- $*\alpha$: *at previous moment α was satisfied.*

Semantics of PDL

A Kripke structure for the language $\mathcal{L}$ of PDL is a system

$$\Sigma = (T, <, m)$$

where

- T – a denumerable set of *timepoints*
- $<$ – a binary relation on T of *temporal ordering*
- $m : T \times Var \rightarrow \{0, 1\}$ — *valuation* of variables at timepoints.

Semantics of PDL

A Kripke structure for the language $\mathcal{L}$ of PDL is a system

$$\Sigma = (T, <, m)$$

where

- T – a denumerable set of *timepoints*
- $<$ – a binary relation on T of *temporal ordering*
- $m : T \times Var \rightarrow \{0, 1\}$ — *valuation* of variables at timepoints.

Typically it is assumed that $<$ is a *strict partial ordering*, i.e.

- *irreflexive*: $t \not< t$ for every $t \in T$
- *antisymmetric*: $t_1 < t_2 \ \& \ t_2 < t_1 \implies t_1 = t_2$ for all $t_1, t_2 \in T$
- *transitive*: $(t_1 < t_2) \ \& \ (t_2 < t_3) \implies (t_1 < t_3)$ for all $t_1, t_2, t_3 \in T$.

Valuation of formulas

Notation:

$$\begin{aligned} succ(t) &= \{t' \in T : (t < t') \ \& \ (\forall t'' \in T \ t < t'' \implies t' < t'')\} \\ prev(t) &= \{t' \in T : (t' < t) \ \& \ (\forall t'' \in T : t'' < t \implies t'' < t')\}. \end{aligned}$$

Valuation of formulas

Notation:

$$\begin{aligned} \text{succ}(t) &= \{t' \in T : (t < t') \ \& \ (\forall t'' \in T \ t < t'' \implies t' < t'')\} \\ \text{prev}(t) &= \{t' \in T : (t' < t) \ \& \ (\forall t'' \in T : t'' < t \implies t'' < t')\}. \end{aligned}$$

$\Sigma, t \models \alpha$ — α is valid at timepoint t in the structure Σ . Inductively, it is defined as follows:

$$\bullet \ \Sigma, t \models p \text{ iff } m(p, t) = 1$$

Valuation of formulas

Notation:

$$succ(t) = \{t' \in T : (t < t') \ \& \ (\forall t'' \in T \ t < t'' \implies t' < t'')\}$$

$$prev(t) = \{t' \in T : (t' < t) \ \& \ (\forall t'' \in T : t'' < t \implies t'' < t')\}.$$

$\Sigma, t \models \alpha$ — α is valid at timepoint t in the structure Σ . Inductively, it is defined as follows:

- $\Sigma, t \models p$ iff $m(p, t) = 1$
- $\Sigma, t \models \neg\alpha$ iff $\Sigma, t \not\models \alpha$

Valuation of formulas

Notation:

$$succ(t) = \{t' \in T : (t < t') \ \& \ (\forall t'' \in T \ t < t'' \implies t' < t'')\}$$

$$prev(t) = \{t' \in T : (t' < t) \ \& \ (\forall t'' \in T : t'' < t \implies t'' < t')\}.$$

$\Sigma, t \models \alpha$ — α is valid at timepoint t in the structure Σ . Inductively, it is defined as follows:

- $\Sigma, t \models p$ iff $m(p, t) = 1$
- $\Sigma, t \models \neg\alpha$ iff $\Sigma, t \not\models \alpha$
- $\Sigma, t \models \alpha \rightarrow \beta$ iff $\Sigma \not\models \alpha$ or $\Sigma, t \models \beta$

Valuation of formulas

Notation:

$$succ(t) = \{t' \in T : (t < t') \ \& \ (\forall t'' \in T \ t < t'' \implies t' < t'')\}$$

$$prev(t) = \{t' \in T : (t' < t) \ \& \ (\forall t'' \in T : t'' < t \implies t'' < t')\}.$$

$\Sigma, t \models \alpha$ — α is valid at timepoint t in the structure Σ . Inductively, it is defined as follows:

- $\Sigma, t \models p$ iff $m(p, t) = 1$
- $\Sigma, t \models \neg\alpha$ iff $\Sigma, t \not\models \alpha$
- $\Sigma, t \models \alpha \rightarrow \beta$ iff $\Sigma \not\models \alpha$ or $\Sigma, t \models \beta$
- $\Sigma, t \models \bigcirc\alpha$ iff $(\forall t' \in T) \ t' \in succ(t) \implies \Sigma, t' \models \alpha$

Valuation of formulas

Notation:

$$succ(t) = \{t' \in T : (t < t') \ \& \ (\forall t'' \in T : t < t'' \implies t' < t'')\}$$

$$prev(t) = \{t' \in T : (t' < t) \ \& \ (\forall t'' \in T : t'' < t \implies t'' < t')\}.$$

$\Sigma, t \models \alpha$ — α is valid at timepoint t in the structure Σ . Inductively, it is defined as follows:

- $\Sigma, t \models p$ iff $m(p, t) = 1$
- $\Sigma, t \models \neg\alpha$ iff $\Sigma, t \not\models \alpha$
- $\Sigma, t \models \alpha \rightarrow \beta$ iff $\Sigma \not\models \alpha$ or $\Sigma, t \models \beta$
- $\Sigma, t \models \bigcirc\alpha$ iff $(\forall t' \in T) t' \in succ(t) \implies \Sigma, t' \models \alpha$
- $\Sigma, t \models *\alpha$ iff $(\forall t' \in T) t' \in prev(t) \implies \Sigma, t' \models \alpha$.

Valuation of formulas (cont.)

- $\Sigma, t \models G\alpha$ iff $(\forall t' \in T) (t < t') \implies \Sigma, t' \models \alpha$

Valuation of formulas (cont.)

- $\Sigma, t \models G\alpha$ iff $(\forall t' \in T) (t < t') \implies \Sigma, t' \models \alpha$
- $\Sigma, t \models F\alpha$ iff $(\exists t' \in T) (t < t') \ \& \ \Sigma, t' \models \alpha$

Valuation of formulas (cont.)

- $\Sigma, t \models G\alpha$ iff $(\forall t' \in T) (t < t') \implies \Sigma, t' \models \alpha$
- $\Sigma, t \models F\alpha$ iff $(\exists t' \in T) (t < t') \ \& \ \Sigma, t' \models \alpha$
- $\Sigma, t \models H\alpha$ iff $(\forall t' \in T) (t' < t) \implies \Sigma, t' \models \alpha$

Valuation of formulas (cont.)

- $\Sigma, t \models G\alpha$ iff $(\forall t' \in T) (t < t') \implies \Sigma, t' \models \alpha$
- $\Sigma, t \models F\alpha$ iff $(\exists t' \in T) (t < t') \ \& \ \Sigma, t' \models \alpha$
- $\Sigma, t \models H\alpha$ iff $(\forall t' \in T) (t' < t) \implies \Sigma, t' \models \alpha$
- $\Sigma, t \models P\alpha$ iff $(\exists t' \in T) (t' < t) \ \& \ \Sigma, t' \models \alpha$

Propositional Dynamic Logic PDL

Syntax of PDL

The alphabet of a language of PDL is that of PL enriched by:

- $\mathcal{At}$ — denumerable set of *atomic actions*
- $?, +, ;, *$ — *action constructors*
- $[], \langle \rangle$ — modal operators.

Syntax of PDL

The alphabet of a language of PDL is that of PL enriched by:

- $\mathcal{At}$ — denumerable set of *atomic actions*
- $?, +, ;, *$ — *action constructors*
- $[], \langle \rangle$ — modal operators.

Basing on $\mathcal{At}$ we build *compound actions* (*actions*, for short) using action constructors. The set of all actions is denoted by $\mathcal{A}$.

Syntax of PDL

The alphabet of a language of PDL is that of PL enriched by:

- $\mathcal{At}$ — denumerable set of **atomic actions**
- $?, +, ;, *$ — **action constructors**
- $[], \langle \rangle$ — modal operators.

Basing on $\mathcal{At}$ we build **compound actions** (**actions**, for short) using action constructors. The set of all actions is denoted by $\mathcal{A}$.

The language $\mathcal{L}$ of PDL is the smallest set of expressions such that

- $Var \subseteq \mathcal{L}$
- $\alpha, \beta \in \mathcal{L} \ \& \ a \in \mathcal{A} \implies \neg\alpha, \alpha \rightarrow \beta, [a]\alpha, \langle a \rangle\alpha \in \mathcal{L}.$

Syntax of PDL

The alphabet of a language of PDL is that of PL enriched by:

- $\mathcal{At}$ — denumerable set of **atomic actions**
- $?, +, ;, *$ — **action constructors**
- $[], \langle \rangle$ — modal operators.

Basing on $\mathcal{At}$ we build **compound actions** (**actions**, for short) using action constructors. The set of all actions is denoted by $\mathcal{A}$.

The language $\mathcal{L}$ of PDL is the smallest set of expressions such that

- $Var \subseteq \mathcal{L}$
- $\alpha, \beta \in \mathcal{L} \ \& \ a \in \mathcal{A} \implies \neg\alpha, \alpha \rightarrow \beta, [a]\alpha, \langle a \rangle \alpha \in \mathcal{L}.$

Intuitively, for every $a \in \mathcal{A}$ and for every $\alpha \in \mathcal{L}$:

- $[a]\alpha$: “if a is performed, then after its execution α always holds”
- $\langle a \rangle \alpha$: “the action a is performed and after its execution α may hold”.

Syntax of PDL (cont.)

The set of actions is the smallest set of expressions such that

- $\mathcal{A}t \subset \mathcal{A}$
- if $a, b \in \mathcal{A}$ and $\alpha \in \mathcal{L}$, then
 - $a; b \in \mathcal{A}$
 - $a + b \in \mathcal{A}$
 - $\alpha? : a, b \in \mathcal{A}$ (conditional action)
 - $a* : \alpha \in \mathcal{A}$ (iteration)

Syntax of PDL (cont.)

The set of actions is the smallest set of expressions such that

- $At \subset A$
- if $a, b \in \mathcal{A}$ and $\alpha \in \mathcal{L}$, then
 - $a; b \in \mathcal{A}$
 - $a + b \in \mathcal{A}$
 - $\alpha? : a, b \in \mathcal{A}$ (conditional action)
 - $a* : \alpha \in \mathcal{A}$ (iteration)

Intuitively:

- $a; b$ — the sequence of actions
- $a + b$ — nondeterministic choice (either a or b is performed)
- $\alpha : a, b$ — if α then a else b
- $a* : \alpha$ — while $\neg\alpha$ do a .

Semantics of PDL

The *Kripke structure* for the language $\mathcal{L}$ of PDL is a system $\Sigma = (S, R, m)$, where

- $S \neq \emptyset$ is a set of *possible states*
- $R : \mathcal{A}t \rightarrow 2^{S \times S}$; for $a \in \mathcal{A}t$, $R(a)$ is a set of pairs (s_1, s_2) of states such that the performance of a in s_1 leads to s_2
- $m : Var \times S \rightarrow \{0, 1\}$ — valuation of variables at states.

Semantics of PDL

The *Kripke structure* for the language $\mathcal{L}$ of PDL is a system $\Sigma = (S, R, m)$, where

- $S \neq \emptyset$ is a set of *possible states*
- $R : \mathcal{At} \rightarrow 2^{S \times S}$; for $a \in \mathcal{At}$, $R(a)$ is a set of pairs (s_1, s_2) of states such that the performance of a in s_1 leads to s_2
- $m : Var \times S \rightarrow \{0, 1\}$ — valuation of variables at states.

Given a structure $\Sigma = (S, R, m)$ and a state $s \in S$, $\Sigma, s \models \alpha$ stands for:
“ α holds in Σ wrt s ”.

Semantics of PDL (cont.)

The mapping $R : \mathcal{At} \rightarrow 2^{S \times S}$ is generalised into the mapping $R^* : \mathcal{A} \rightarrow 2^{S \times S}$ as follows:

- $R^*(a) = R(a)$ for every $a \in \mathcal{At}$

Semantics of PDL (cont.)

The mapping $R : \mathcal{At} \rightarrow 2^{S \times S}$ is generalised into the mapping $R^* : \mathcal{A} \rightarrow 2^{S \times S}$ as follows:

- $R^*(a) = R(a)$ for every $a \in \mathcal{At}$
- $(s, s') \in R^*(a; b)$ iff $(\exists s'' \in S) (s, s'') \in R^*(a) \ \& \ (s'', s') \in R^*(b)$

Semantics of PDL (cont.)

The mapping $R : \mathcal{At} \rightarrow 2^{S \times S}$ is generalised into the mapping $R^* : \mathcal{A} \rightarrow 2^{S \times S}$ as follows:

- $R^*(a) = R(a)$ for every $a \in \mathcal{At}$
- $(s, s') \in R^*(a; b)$ iff $(\exists s'' \in S) (s, s'') \in R^*(a) \ \& \ (s'', s') \in R^*(b)$
- $R^*(a + b) = R^*(a) \cup R^*(b)$

Semantics of PDL (cont.)

The mapping $R : \mathcal{At} \rightarrow 2^{S \times S}$ is generalised into the mapping $R^* : \mathcal{A} \rightarrow 2^{S \times S}$ as follows:

- $R^*(a) = R(a)$ for every $a \in \mathcal{At}$
- $(s, s') \in R^*(a; b)$ iff $(\exists s'' \in S) (s, s'') \in R^*(a) \ \& \ (s'', s') \in R^*(b)$
- $R^*(a + b) = R^*(a) \cup R^*(b)$
- $(s, s') \in R^*(\alpha? : a, b)$ iff
 - $\Sigma, s \models \alpha \implies (s, s') \in R^*(a)$
 - $\Sigma, s \not\models \alpha \implies (s, s') \in R^*(b)$

Semantics of PDL (cont.)

The mapping $R : \mathcal{At} \rightarrow 2^{S \times S}$ is generalised into the mapping $R^* : \mathcal{A} \rightarrow 2^{S \times S}$ as follows:

- $R^*(a) = R(a)$ for every $a \in \mathcal{At}$
- $(s, s') \in R^*(a; b)$ iff $(\exists s'' \in S) (s, s'') \in R^*(a) \ \& \ (s'', s') \in R^*(b)$
- $R^*(a + b) = R^*(a) \cup R^*(b)$
- $(s, s') \in R^*(\alpha? : a, b)$ iff
 - $\Sigma, s \models \alpha \implies (s, s') \in R^*(a)$
 - $\Sigma, s \not\models \alpha \implies (s, s') \in R^*(b)$
- $(s, s') \in R^*(a^* : \alpha)$ iff $(\exists n \in \mathbb{N}) (\exists s_0, s_1, \dots, s_n \in S)$
 - $s_0 = s$ and $s_n = s'$
 - $(\forall i \in \mathbb{N}) \ 0 \leq i < n \implies (s_i, s_{i+1}) \in R^*(a)$
 - $(\forall i \in \mathbb{N}) \ 0 \leq i < n \implies \Sigma, s_i \models \neg \alpha$
 - $\Sigma, s' \models \alpha$.

Semantics of PDL (cont.)

The mapping $R : \mathcal{At} \rightarrow 2^{S \times S}$ is generalised into the mapping $R^* : \mathcal{A} \rightarrow 2^{S \times S}$ as follows:

- $R^*(a) = R(a)$ for every $a \in \mathcal{At}$
- $(s, s') \in R^*(a; b)$ iff $(\exists s'' \in S) (s, s'') \in R^*(a) \ \& \ (s'', s') \in R^*(b)$
- $R^*(a + b) = R^*(a) \cup R^*(b)$
- $(s, s') \in R^*(\alpha? : a, b)$ iff
 - $\Sigma, s \models \alpha \implies (s, s') \in R^*(a)$
 - $\Sigma, s \not\models \alpha \implies (s, s') \in R^*(b)$
- $(s, s') \in R^*(a^* : \alpha)$ iff $(\exists n \in \mathbb{N}) (\exists s_0, s_1, \dots, s_n \in S)$
 - $s_0 = s$ and $s_n = s'$
 - $(\forall i \in \mathbb{N}) \ 0 \leq i < n \implies (s_i, s_{i+1}) \in R^*(a)$
 - $(\forall i \in \mathbb{N}) \ 0 \leq i < n \implies \Sigma, s_i \models \neg \alpha$
 - $\Sigma, s' \models \alpha$.

Semantics of PDL (cont.)

The mapping $R : \mathcal{At} \rightarrow 2^{S \times S}$ is generalised into the mapping $R^* : \mathcal{A} \rightarrow 2^{S \times S}$ as follows:

- $R^*(a) = R(a)$ for every $a \in \mathcal{At}$
- $(s, s') \in R^*(a; b)$ iff $(\exists s'' \in S) (s, s'') \in R^*(a) \ \& \ (s'', s') \in R^*(b)$
- $R^*(a + b) = R^*(a) \cup R^*(b)$
- $(s, s') \in R^*(\alpha? : a, b)$ iff
 - $\Sigma, s \models \alpha \implies (s, s') \in R^*(a)$
 - $\Sigma, s \not\models \alpha \implies (s, s') \in R^*(b)$
- $(s, s') \in R^*(a^* : \alpha)$ iff $(\exists n \in \mathbb{N}) (\exists s_0, s_1, \dots, s_n \in S)$
 - $s_0 = s$ and $s_n = s'$
 - $(\forall i \in \mathbb{N}) \ 0 \leq i < n \implies (s_i, s_{i+1}) \in R^*(a)$
 - $(\forall i \in \mathbb{N}) \ 0 \leq i < n \implies \Sigma, s_i \models \neg \alpha$
 - $\Sigma, s' \models \alpha$.

Semantics of PDL (cont.)

The mapping $R : \mathcal{At} \rightarrow 2^{S \times S}$ is generalised into the mapping $R^* : \mathcal{A} \rightarrow 2^{S \times S}$ as follows:

- $R^*(a) = R(a)$ for every $a \in \mathcal{At}$
- $(s, s') \in R^*(a; b)$ iff $(\exists s'' \in S) (s, s'') \in R^*(a) \ \& \ (s'', s') \in R^*(b)$
- $R^*(a + b) = R^*(a) \cup R^*(b)$
- $(s, s') \in R^*(\alpha? : a, b)$ iff
 - $\Sigma, s \models \alpha \implies (s, s') \in R^*(a)$
 - $\Sigma, s \not\models \alpha \implies (s, s') \in R^*(b)$
- $(s, s') \in R^*(a^* : \alpha)$ iff $(\exists n \in \mathbb{N}) (\exists s_0, s_1, \dots, s_n \in S)$
 - $s_0 = s$ and $s_n = s'$
 - $(\forall i \in \mathbb{N}) \ 0 \leq i < n \implies (s_i, s_{i+1}) \in R^*(a)$
 - $(\forall i \in \mathbb{N}) \ 0 \leq i < n \implies \Sigma, s_i \models \neg \alpha$
 - $\Sigma, s' \models \alpha$.

Semantics of PDL (cont.)

The mapping $R : \mathcal{At} \rightarrow 2^{S \times S}$ is generalised into the mapping $R^* : \mathcal{A} \rightarrow 2^{S \times S}$ as follows:

- $R^*(a) = R(a)$ for every $a \in \mathcal{At}$
- $(s, s') \in R^*(a; b)$ iff $(\exists s'' \in S) (s, s'') \in R^*(a) \ \& \ (s'', s') \in R^*(b)$
- $R^*(a + b) = R^*(a) \cup R^*(b)$
- $(s, s') \in R^*(\alpha? : a, b)$ iff
 - $\Sigma, s \models \alpha \implies (s, s') \in R^*(a)$
 - $\Sigma, s \not\models \alpha \implies (s, s') \in R^*(b)$
- $(s, s') \in R^*(a^* : \alpha)$ iff $(\exists n \in \mathbb{N}) (\exists s_0, s_1, \dots, s_n \in S)$
 - $s_0 = s$ and $s_n = s'$
 - $(\forall i \in \mathbb{N}) \ 0 \leq i < n \implies (s_i, s_{i+1}) \in R^*(a)$
 - $(\forall i \in \mathbb{N}) \ 0 \leq i < n \implies \Sigma, s_i \models \neg \alpha$
 - $\Sigma, s' \models \alpha$.

Semantics of PDL (cont.)

The mapping $R : \mathcal{At} \rightarrow 2^{S \times S}$ is generalised into the mapping $R^* : \mathcal{A} \rightarrow 2^{S \times S}$ as follows:

- $R^*(a) = R(a)$ for every $a \in \mathcal{At}$
- $(s, s') \in R^*(a; b)$ iff $(\exists s'' \in S) (s, s'') \in R^*(a) \ \& \ (s'', s') \in R^*(b)$
- $R^*(a + b) = R^*(a) \cup R^*(b)$
- $(s, s') \in R^*(\alpha? : a, b)$ iff
 - $\Sigma, s \models \alpha \implies (s, s') \in R^*(a)$
 - $\Sigma, s \not\models \alpha \implies (s, s') \in R^*(b)$
- $(s, s') \in R^*(a^* : \alpha)$ iff $(\exists n \in \mathbb{N}) (\exists s_0, s_1, \dots, s_n \in S)$
 - $s_0 = s$ and $s_n = s'$
 - $(\forall i \in \mathbb{N}) \ 0 \leq i < n \implies (s_i, s_{i+1}) \in R^*(a)$
 - $(\forall i \in \mathbb{N}) \ 0 \leq i < n \implies \Sigma, s_i \models \neg \alpha$
 - $\Sigma, s' \models \alpha$.

Semantics of PDL (cont.)

The mapping $R : \mathcal{At} \rightarrow 2^{S \times S}$ is generalised into the mapping $R^* : \mathcal{A} \rightarrow 2^{S \times S}$ as follows:

- $R^*(a) = R(a)$ for every $a \in \mathcal{At}$
- $(s, s') \in R^*(a; b)$ iff $(\exists s'' \in S) (s, s'') \in R^*(a) \ \& \ (s'', s') \in R^*(b)$
- $R^*(a + b) = R^*(a) \cup R^*(b)$
- $(s, s') \in R^*(\alpha? : a, b)$ iff
 - $\Sigma, s \models \alpha \implies (s, s') \in R^*(a)$
 - $\Sigma, s \not\models \alpha \implies (s, s') \in R^*(b)$
- $(s, s') \in R^*(a^* : \alpha)$ iff $(\exists n \in \mathbb{N}) (\exists s_0, s_1, \dots, s_n \in S)$
 - $s_0 = s$ and $s_n = s'$
 - $(\forall i \in \mathbb{N}) \ 0 \leq i < n \implies (s_i, s_{i+1}) \in R^*(a)$
 - $(\forall i \in \mathbb{N}) \ 0 \leq i < n \implies \Sigma, s_i \models \neg \alpha$
 - $\Sigma, s' \models \alpha$.

Valuation of formulas in PDL

Let $\Sigma = (S, R, m)$ and $s \in S$.

• $\Sigma, s \models p$ iff $m(s, p) = 1$

Valuation of formulas in PDL

Let $\Sigma = (S, R, m)$ and $s \in S$.

• $\Sigma, s \models p$ iff $m(s, p) = 1$

• $\Sigma, s \models \neg\alpha$ iff $\Sigma, s \not\models \alpha$

Valuation of formulas in PDL

Let $\Sigma = (S, R, m)$ and $s \in S$.

- $\Sigma, s \models p$ iff $m(s, p) = 1$
- $\Sigma, s \models \neg\alpha$ iff $\Sigma, s \not\models \alpha$
- $\Sigma, s \models \alpha \rightarrow \beta$ iff $\Sigma, s \models \neg\alpha$ & $\Sigma, s \models \beta$

Valuation of formulas in PDL

Let $\Sigma = (S, R, m)$ and $s \in S$.

- $\Sigma, s \models p$ iff $m(s, p) = 1$
- $\Sigma, s \models \neg\alpha$ iff $\Sigma, s \not\models \alpha$
- $\Sigma, s \models \alpha \rightarrow \beta$ iff $\Sigma, s \models \neg\alpha$ & $\Sigma, s \models \beta$
- $\Sigma, s \models [a]\alpha$ iff $(\forall s' \in S) (s, s') \in R^*(a) \implies \Sigma, s' \models \alpha$

Valuation of formulas in PDL

Let $\Sigma = (S, R, m)$ and $s \in S$.

- $\Sigma, s \models p$ iff $m(s, p) = 1$
- $\Sigma, s \models \neg\alpha$ iff $\Sigma, s \not\models \alpha$
- $\Sigma, s \models \alpha \rightarrow \beta$ iff $\Sigma, s \models \neg\alpha$ & $\Sigma, s \models \beta$
- $\Sigma, s \models [a]\alpha$ iff $(\forall s' \in S) (s, s') \in R^*(a) \implies \Sigma, s' \models \alpha$
- $\Sigma, s \models \langle a \rangle \alpha$ iff $(\exists s' \in S) (s, s') \in R^*(a) \text{ \& } \Sigma, s' \models \alpha$.

Belief–Desires–Intentions Structures (BDI)

BDI systems

BDI systems are logical systems for modeling agents, their knowledge, beliefs, intentions, and actions.

BDI systems

BDI systems are logical systems for modeling agents, their knowledge, beliefs, intentions, and actions.

Agent's knowledge, beliefs and desires are modeled by means of the suitable binary relations on the set of possible states. Specifically, for every agent a , we have three relations $K, B, D \subseteq S \times S$ viewed as:

- $(s, s') \in K(a)$ iff s' is K-accessible from s for a (wrt his knowledge)
- $(s, s') \in B(a)$ iff s' is B-accessible from s for a (wrt his beliefs)
- $(s, s') \in D(a)$ iff s' is D-accessible from s for a (wrt his desires).

BDI systems (cont.)

Typically, it is assumed that

• the relation B is

- *serial*: $(\forall s \in S) (\exists s' \in S) (s, s') \in B(a)$
- *Euclidean*: $(\forall s, s', s'' \in S)$
 $(s, s') \in B(a) \ \& \ (s, s'') \in B(a) \implies (s', s'') \in B(a)$
- *symmetric*: $(\forall s, s' \in S) (s, s') \in B(a) \implies (s', s) \in B(a).$

BDI systems (cont.)

Typically, it is assumed that

- the relation B is
 - *serial*: $(\forall s \in S) (\exists s' \in S) (s, s') \in B(a)$
 - *Euclidean*: $(\forall s, s', s'' \in S)$
 $(s, s') \in B(a) \ \& \ (s, s'') \in B(a) \implies (s', s'') \in B(a)$
 - *symmetric*: $(\forall s, s' \in S) (s, s') \in B(a) \implies (s', s) \in B(a)$.
- the relation D is serial only.

BDI systems (cont.)

Typically, it is assumed that

- the relation B is
 - *serial*: $(\forall s \in S) (\exists s' \in S) (s, s') \in B(a)$
 - *Euclidean*: $(\forall s, s', s'' \in S)$
 $(s, s') \in B(a) \ \& \ (s, s'') \in B(a) \implies (s', s'') \in B(a)$
 - *symmetric*: $(\forall s, s' \in S) (s, s') \in B(a) \implies (s', s) \in B(a)$.
- the relation D is serial only.

For modeling of agent's intentions there are two approaches:

- yet another relation $I(a) \subseteq S \times S$: *α is a 's intention iff it holds in all states I -accessible from s for a*
- *α is a 's intention in $s \in S$ iff it holds at some point on each path started in s .*

Syntax of BDI

An alphabet of a language of BDI consists of:

- a non-empty denumerable set Var of *propositional variables*

Syntax of BDI

An alphabet of a language of BDI consists of:

- a non-empty denumerable set Var of *propositional variables*
- a non-empty denumerable set Ag of *agents*

Syntax of BDI

An alphabet of a language of BDI consists of:

- a non-empty denumerable set Var of *propositional variables*
- a non-empty denumerable set Ag of *agents*
- a non-empty denumerable set At of *atomic actions*

Syntax of BDI

An alphabet of a language of BDI consists of:

- a non-empty denumerable set Var of *propositional variables*
- a non-empty denumerable set Ag of *agents*
- a non-empty denumerable set At of *atomic actions*
- logical connectives $\neg, \rightarrow$

Syntax of BDI

An alphabet of a language of BDI consists of:

- a non-empty denumerable set Var of **propositional variables**
- a non-empty denumerable set Ag of **agents**
- a non-empty denumerable set At of **atomic actions**
- logical connectives $\neg, \rightarrow$
- modal operators B (**beliefs**) and D (**desires**)

Syntax of BDI

An alphabet of a language of BDI consists of:

- a non-empty denumerable set Var of *propositional variables*
- a non-empty denumerable set Ag of *agents*
- a non-empty denumerable set At of *atomic actions*
- logical connectives $\neg, \rightarrow$
- modal operators B (*beliefs*) and D (*desires*)
- temporal operators $\bigcirc, F$ (*in the future*), G (*going to be*)

Syntax of BDI

An alphabet of a language of BDI consists of:

- a non-empty denumerable set Var of *propositional variables*
- a non-empty denumerable set Ag of *agents*
- a non-empty denumerable set At of *atomic actions*
- logical connectives $\neg, \rightarrow$
- modal operators B (*beliefs*) and D (*desires*)
- temporal operators $\bigcirc, F$ (*in the future*), G (*going to be*)
- dynamic operators $[], \langle \rangle$

Syntax of BDI

An alphabet of a language of BDI consists of:

- a non-empty denumerable set Var of *propositional variables*
- a non-empty denumerable set Ag of *agents*
- a non-empty denumerable set At of *atomic actions*
- logical connectives $\neg, \rightarrow$
- modal operators B (*beliefs*) and D (*desires*)
- temporal operators $\bigcirc, F$ (*in the future*), G (*going to be*)
- dynamic operators $[], \langle \rangle$
- path operators I (*intentions*), E .

Syntax of BDI

An alphabet of a language of BDI consists of:

- a non–empty denumerable set Var of **propositional variables**
- a non–empty denumerable set Ag of **agents**
- a non–empty denumerable set At of **atomic actions**
- logical connectives $\neg, \rightarrow$
- modal operators B (**beliefs**) and D (**desires**)
- temporal operators $\bigcirc, F$ (**in the future**), G (**going to be**)
- dynamic operators $[], \langle \rangle$
- path operators I (**intentions**), E .

Intuitively, $E\alpha$ means: “ α holds on some path started in the state s ”.

BDI formulas

There are 2 types of formulas in BDI languages:

- *path formulas* (subset of the language) — interpreted in stated wrt paths started these states
- *formulas* — interpreted in stated only.

BDI formulas

There are 2 types of formulas in BDI languages:

- *path formulas* (subset of the language) — interpreted in stated wrt paths started these states
- *formulas* — interpreted in stated only.

The set $\mathcal{F}_\pi$ of path formulas is the smallest set of expressions such that

- $Var \subset \mathcal{F}_\pi$
- if $\alpha, \beta \in \mathcal{F}_\pi$, $a \in \mathcal{At}$, and $x \in \mathcal{Ag}$, then
$$\neg\alpha, \alpha \rightarrow \beta, \bigcirc\alpha, G\alpha, F\alpha, x[a]\alpha, x\langle a\rangle\alpha \in \mathcal{F}_\pi.$$

BDI formulas

There are 2 types of formulas in BDI languages:

- *path formulas* (subset of the language) — interpreted in stated wrt paths started these states
- *formulas* — interpreted in stated only.

The set $\mathcal{F}_\pi$ of path formulas is the smallest set of expressions such that

- $Var \subset \mathcal{F}_\pi$
- if $\alpha, \beta \in \mathcal{F}_\pi$, $a \in \mathcal{At}$, and $x \in \mathcal{Ag}$, then
$$\neg\alpha, \alpha \rightarrow \beta, \bigcirc\alpha, G\alpha, F\alpha, x[a]\alpha, x\langle a\rangle\alpha \in \mathcal{F}_\pi.$$

The set $\mathcal{L}$ of all formulas is the smallest set of expressions such that

- $\mathcal{F}_\pi \subset \mathcal{L}$
- $\alpha \in \mathcal{L}, x \in \mathcal{Ag} \implies B(x, \alpha), D(x, \alpha), E(x, \alpha), I(x, \alpha) \in \mathcal{L}.$

Semantics of BDI — notations

Notation:

- $S^* = \{(s_1, s_2, \dots) : s_i \in S \text{ \& } s_i < s_{i+1} \text{ for every } i \in N\}$,
i.e. S^* is the set of all sequences (finite, infinite) of states.

Semantics of BDI — notations

Notation:

- $S^* = \{(s_1, s_2, \dots,) : s_i \in S \text{ \& } s_i < s_{i+1} \text{ for every } i \in N\}$,
i.e. S^* is the set of all sequences (finite, infinite) of states.
- S_s^* – the set of paths starting in the state s .

Semantics of BDI — notations

Notation:

- $S^* = \{(s_1, s_2, \dots) : s_i \in S \ \& \ s_i < s_{i+1} \text{ for every } i \in N\}$,
i.e. S^* is the set of all sequences (finite, infinite) of states.
- S_s^* – the set of paths starting in the state s .
- for $s \in S$ and $\wp \in S^*$, $s' \in succ(\wp, s)$ iff
 $(s' \in \wp) \ \& \ (s < s') \ \& \ (\forall s'' \in S) (s'' \in \wp) \ \& \ (s < s'') \implies (s'' \not< s')$.

Semantics of BDI

A *Kripke structure* for the language $\mathcal{L}$ of BDI is a system

$$\Sigma = (S, <, B, D, I, m, \mu)$$

such that

- $S \neq \emptyset$ is a set of states

Semantics of BDI

A *Kripke structure* for the language $\mathcal{L}$ of BDI is a system

$$\Sigma = (S, <, B, D, I, m, \mu)$$

such that

- $S \neq \emptyset$ is a set of states
- $< \subseteq S \times S$ is a temporal ordering

Semantics of BDI

A *Kripke structure* for the language $\mathcal{L}$ of BDI is a system

$$\Sigma = (S, <, B, D, I, m, \mu)$$

such that

- $S \neq \emptyset$ is a set of states
- $< \subseteq S \times S$ is a temporal ordering
- $B, D : Ag \rightarrow 2^{S \times S}; B \subseteq D$ (*an agent's desires are his beliefs*)

Semantics of BDI

A *Kripke structure* for the language $\mathcal{L}$ of BDI is a system

$$\Sigma = (S, <, B, D, I, m, \mu)$$

such that

- $S \neq \emptyset$ is a set of states
- $< \subseteq S \times S$ is a temporal ordering
- $B, D : Ag \rightarrow 2^{S \times S}$; $B \subseteq D$ (*an agent's desires are his beliefs*)
- $I : Ag \times S \rightarrow 2^{S^*}$ — for $x \in Ag$ and for $s \in S$, $I(x, s)$ is the set of “preferable paths” for x in s ; it is assumed that $I(x, s) \neq \emptyset$ and for every $\wp \in I(x, s)$, $\wp \in S_s^*$

Semantics of BDI

A *Kripke structure* for the language $\mathcal{L}$ of BDI is a system

$$\Sigma = (S, <, B, D, I, m, \mu)$$

such that

- $S \neq \emptyset$ is a set of states
- $< \subseteq S \times S$ is a temporal ordering
- $B, D : Ag \rightarrow 2^{S \times S}$; $B \subseteq D$ (*an agent's desires are his beliefs*)
- $I : Ag \times S \rightarrow 2^{S^*}$ — for $x \in Ag$ and for $s \in S$, $I(x, s)$ is the set of “preferable paths” for x in s ; it is assumed that $I(x, s) \neq \emptyset$ and for every $\wp \in I(x, s)$, $\wp \in S_s^*$
- $m : Var \times S \rightarrow \{0, 1\}$

Semantics of BDI

A *Kripke structure* for the language $\mathcal{L}$ of BDI is a system

$$\Sigma = (S, <, B, D, I, m, \mu)$$

such that

- $S \neq \emptyset$ is a set of states
- $< \subseteq S \times S$ is a temporal ordering
- $B, D : Ag \rightarrow 2^{S \times S}$; $B \subseteq D$ (*an agent's desires are his beliefs*)
- $I : Ag \times S \rightarrow 2^{S^*}$ — for $x \in Ag$ and for $s \in S$, $I(x, s)$ is the set of “preferable paths” for x in s ; it is assumed that $I(x, s) \neq \emptyset$ and for every $\wp \in I(x, s)$, $\wp \in S_s^*$
- $m : Var \times S \rightarrow \{0, 1\}$
- $\mu : Ag \times \mathcal{At} \rightarrow 2^{S \times S}$; for $x \in Ag$ and $a \in \mathcal{At}$, $\mu(x, s)$ is the set of all pairs (s, s') such that performing a in s by x leads to s' .

Semantics of BDI

A *Kripke structure* for the language $\mathcal{L}$ of BDI is a system

$$\Sigma = (S, <, B, D, I, m, \mu)$$

such that

- $S \neq \emptyset$ is a set of states
- $< \subseteq S \times S$ is a temporal ordering
- $B, D : Ag \rightarrow 2^{S \times S}$; $B \subseteq D$ (an agent's desires are his beliefs)
- $I : Ag \times S \rightarrow 2^{S^*}$ — for $x \in Ag$ and for $s \in S$, $I(x, s)$ is the set of “preferable paths” for x in s ; it is assumed that $I(x, s) \neq \emptyset$ and for every $\wp \in I(x, s)$, $\wp \in S_s^*$
- $m : Var \times S \rightarrow \{0, 1\}$
- $\mu : Ag \times \mathcal{A}t \rightarrow 2^{S \times S}$; for $x \in Ag$ and $a \in \mathcal{A}t$, $\mu(x, s)$ is the set of all pairs (s, s') such that performing a in s by x leads to s' .

Semantics of BDI

Let $\Sigma = (S, <, B, D, I, m, \mu)$, $s \in S$, $\wp \in S^*$.

• $\Sigma, \wp, s \models p$ iff $(s \in \wp) \ \& \ \Sigma, s \models p$ for every $p \in Var$

Semantics of BDI

Let $\Sigma = (S, <, B, D, I, m, \mu)$, $s \in S$, $\wp \in S^*$.

- $\Sigma, \wp, s \models p$ iff $(s \in \wp) \ \& \ \Sigma, s \models p$ for every $p \in Var$
- $\Sigma, \wp, s \models \bigcirc \alpha$ iff $(s \in \wp) \ \& \ \Sigma, succ(\wp, s) \models \alpha$

Semantics of BDI

Let $\Sigma = (S, <, B, D, I, m, \mu)$, $s \in S$, $\wp \in S^*$.

- $\Sigma, \wp, s \models p$ iff $(s \in \wp) \ \& \ \Sigma, s \models p$ for every $p \in Var$
- $\Sigma, \wp, s \models \bigcirc \alpha$ iff $(s \in \wp) \ \& \ \Sigma, succ(\wp, s) \models \alpha$
- $\Sigma, \wp, s \models F\alpha$ iff $(s \in \wp) \ \& \ (\exists s' \in S) \ s' \in \wp \ \& \ s < s' \ \& \ \Sigma, \wp, s' \models \alpha$

Semantics of BDI

Let $\Sigma = (S, <, B, D, I, m, \mu)$, $s \in S$, $\wp \in S^*$.

- $\Sigma, \wp, s \models p$ iff $(s \in \wp) \ \& \ \Sigma, s \models p$ for every $p \in Var$
- $\Sigma, \wp, s \models \bigcirc \alpha$ iff $(s \in \wp) \ \& \ \Sigma, succ(\wp, s) \models \alpha$
- $\Sigma, \wp, s \models F\alpha$ iff $(s \in \wp) \ \& \ (\exists s' \in S) \ s' \in \wp \ \& \ s < s' \ \& \ \Sigma, \wp, s' \models \alpha$
- $\Sigma, \wp, s \models G\alpha$ iff $(s \in \wp) \ \& \ (\forall s' \in S) \ s' \in \wp \ \& \ s < s' \implies \Sigma, \wp, s' \models \alpha$

Semantics of BDI

Let $\Sigma = (S, <, B, D, I, m, \mu)$, $s \in S$, $\wp \in S^*$.

- $\Sigma, \wp, s \models p$ iff $(s \in \wp) \ \& \ \Sigma, s \models p$ for every $p \in Var$
- $\Sigma, \wp, s \models \bigcirc \alpha$ iff $(s \in \wp) \ \& \ \Sigma, succ(\wp, s) \models \alpha$
- $\Sigma, \wp, s \models F\alpha$ iff $(s \in \wp) \ \& \ (\exists s' \in S) \ s' \in \wp \ \& \ s < s' \ \& \ \Sigma, \wp, s' \models \alpha$
- $\Sigma, \wp, s \models G\alpha$ iff $(s \in \wp) \ \& \ (\forall s' \in S) \ s' \in \wp \ \& \ s < s' \implies \Sigma, \wp, s' \models \alpha$
- $\Sigma, \wp, s \models x[a]\alpha$ iff $(s \in \wp) \ \& \ (\forall s' \in S) \ s' \in \wp \ \& \ (s, s') \in \mu(x, a) \implies \Sigma, \wp, s' \models \alpha$

Semantics of BDI

Let $\Sigma = (S, <, B, D, I, m, \mu)$, $s \in S$, $\wp \in S^*$.

- $\Sigma, \wp, s \models p$ iff $(s \in \wp) \ \& \ \Sigma, s \models p$ for every $p \in Var$
- $\Sigma, \wp, s \models \bigcirc \alpha$ iff $(s \in \wp) \ \& \ \Sigma, succ(\wp, s) \models \alpha$
- $\Sigma, \wp, s \models F\alpha$ iff $(s \in \wp) \ \& \ (\exists s' \in S) \ s' \in \wp \ \& \ s < s' \ \& \ \Sigma, \wp, s' \models \alpha$
- $\Sigma, \wp, s \models G\alpha$ iff $(s \in \wp) \ \& \ (\forall s' \in S) \ s' \in \wp \ \& \ s < s' \implies \Sigma, \wp, s' \models \alpha$
- $\Sigma, \wp, s \models x[a]\alpha$ iff $(s \in \wp) \ \& \ (\forall s' \in S) \ s' \in \wp \ \& \ (s, s') \in \mu(x, a) \implies \Sigma, \wp, s' \models \alpha$
- $\Sigma, \wp, s \models x\langle a \rangle \alpha$ iff $(s \in \wp) \ \& \ (\exists s' \in S) \ s' \in \wp \ \& \ (s, s') \in \mu(x, a) \implies \Sigma, \wp, s' \models \alpha$.

Semantics of BDI (cont.)

Let $\Sigma = (S, <, B, D, I, m, \mu)$, $x \in Ag$, $s \in S$.

• $\Sigma, s \models p$ iff $m(p, s) = 1$ for every $p \in Var$

Semantics of BDI (cont.)

Let $\Sigma = (S, <, B, D, I, m, \mu)$, $x \in Ag$, $s \in S$.

- $\Sigma, s \models p$ iff $m(p, s) = 1$ for every $p \in Var$
- $\Sigma, s \models B(x, \alpha)$ iff $(\forall s' \in S) s' \in B(x, s) \implies \Sigma, s' \models \alpha$

Semantics of BDI (cont.)

Let $\Sigma = (S, <, B, D, I, m, \mu)$, $x \in Ag$, $s \in S$.

- $\Sigma, s \models p$ iff $m(p, s) = 1$ for every $p \in Var$
- $\Sigma, s \models B(x, \alpha)$ iff $(\forall s' \in S) s' \in B(x, s) \implies \Sigma, s' \models \alpha$
- $\Sigma, s \models D(x, \alpha)$ iff $(\forall s' \in S) s' \in D(x, s) \implies \Sigma, s' \models \alpha$

Semantics of BDI (cont.)

Let $\Sigma = (S, <, B, D, I, m, \mu)$, $x \in Ag$, $s \in S$.

- $\Sigma, s \models p$ iff $m(p, s) = 1$ for every $p \in Var$
- $\Sigma, s \models B(x, \alpha)$ iff $(\forall s' \in S) s' \in B(x, s) \implies \Sigma, s' \models \alpha$
- $\Sigma, s \models D(x, \alpha)$ iff $(\forall s' \in S) s' \in D(x, s) \implies \Sigma, s' \models \alpha$
- $\Sigma, s \models \alpha$ iff $(\exists \wp \in S_s^*) \Sigma, \wp, s \models \alpha$

Semantics of BDI (cont.)

Let $\Sigma = (S, <, B, D, I, m, \mu)$, $x \in Ag$, $s \in S$.

- $\Sigma, s \models p$ iff $m(p, s) = 1$ for every $p \in Var$
- $\Sigma, s \models B(x, \alpha)$ iff $(\forall s' \in S) s' \in B(x, s) \implies \Sigma, s' \models \alpha$
- $\Sigma, s \models D(x, \alpha)$ iff $(\forall s' \in S) s' \in D(x, s) \implies \Sigma, s' \models \alpha$
- $\Sigma, s \models \alpha$ iff $(\exists \wp \in S_s^*) \Sigma, \wp, s \models \alpha$
- $\Sigma, s \models I(x, \alpha)$ iff $(\forall \wp \in S_s^*) \wp \in I(x, s) \implies \Sigma, \wp, s \models F\alpha$.

Properties of B

Remark: Every symmetric and Euclidean relation is transitive.

- $\models B(x, \alpha) \wedge B(x, \alpha \rightarrow \beta) \rightarrow B(x, \beta)$

Properties of B

Remark: Every symmetric and Euclidean relation is transitive.

- $\models B(x, \alpha) \wedge B(x, \alpha \rightarrow \beta) \rightarrow B(x, \beta)$

- $\models B(x, \alpha) \rightarrow \neg B(x, \neg\alpha)$

(follows from seriality of B)

Properties of B

Remark: Every symmetric and Euclidean relation is transitive.

- $\models B(x, \alpha) \wedge B(x, \alpha \rightarrow \beta) \rightarrow B(x, \beta)$

- $\models B(x, \alpha) \rightarrow \neg B(x, \neg\alpha)$

(follows from seriality of B)

- $\models B(x, \alpha) \rightarrow B(x, B(x, \alpha))$

(follows from transitivity of B)

Properties of B

Remark: Every symmetric and Euclidean relation is transitive.

- $\models B(x, \alpha) \wedge B(x, \alpha \rightarrow \beta) \rightarrow B(x, \beta)$
- $\models B(x, \alpha) \rightarrow \neg B(x, \neg \alpha)$
(follows from seriality of B)
- $\models B(x, \alpha) \rightarrow B(x, B(x, \alpha))$
(follows from transitivity of B)
- $\models B(x, \alpha) \rightarrow \neg B(x, \alpha)$
(follows from symmetry and transitivity of B).

Properties of D

- $\models D(x, \alpha) \wedge D(x, \alpha \rightarrow \beta) \rightarrow D(x, \beta)$

Properties of D

- $\models D(x, \alpha) \wedge D(x, \alpha \rightarrow \beta) \rightarrow D(x, \beta)$
- $\models D(x, \alpha) \rightarrow \neg D(x, \neg \alpha)$

Properties of D

- $\models D(x, \alpha) \wedge D(x, \alpha \rightarrow \beta) \rightarrow D(x, \beta)$
- $\models D(x, \alpha) \rightarrow \neg D(x, \neg \alpha)$
- $\models D(x, \alpha) \rightarrow B(x, \alpha).$

Properties of I

- **satisfiability:** $\models I(x, \alpha) \rightarrow EF\alpha$ for every $x \in Ag$.

Properties of I

- **satisfiability:** $\models I(x, \alpha) \rightarrow EF\alpha$ for every $x \in Ag$.
- $\models I(x, \alpha) \wedge I(x, \beta) \rightarrow I(x, F\alpha \wedge F\beta)$

If x has an intention α and he has an intention β , then his intention is to achieve both α and β in some (unspecified) order: firstly α then β , or conversely: first β then α .

Properties of I

- **satisfiability:** $\models I(x, \alpha) \rightarrow EF\alpha$ for every $x \in Ag$.

- $\models I(x, \alpha) \wedge I(x, \beta) \rightarrow I(x, F\alpha \wedge F\beta)$

If x has an intention α and he has an intention β , then his intention is to achieve both α and β in some (unspecified) order: firstly α then β , or conversely: first β then α .

- $EG(I(x, \alpha) \wedge \neg\alpha)$ is satisfiable! This formula states that at some state s there exists a path $\wp$ starting in s such that in every state on $\wp$ both $\neg\alpha$ and $I(x, \alpha)$ hold. Therefore, for every state s' on $\wp$ there is a “preferable” path $\wp'$ (starting in s') such that at some state on $\wp'$ α holds. At the same time, in the state s' α does not hold.

Concluding, it reflects the fact: an agent always has some intentions (to achieve goals he aims to), but not necessarily he achieves these goals.

Thank you for your attention!

Any questions are welcome.